

Dong Payton Pei

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SUMMARY

Bachelor of Computer Science from Carleton University with extensive co-op experience in software testing and development. Skilled in TypeScript, JavaScript, Python, and various other technologies. Seeking a Software Engineer position to leverage my skills in designing, developing, and testing software solutions.

EDUCATION

University of Ottawa <i>Master of Engineering System Science</i>	Ottawa, ON <i>Present</i>
Carleton University <i>Bachelor of Computer Science</i>	Ottawa, ON <i>2017 — 2023</i>

WORK EXPERIENCE

Software Developer Co-op Ford Motor Canada	Apr 2020 — Aug 2020 <i>Ottawa, ON</i>
Participated in Agile development, developed unit test scripts using C++11 Google Test Framework, and utilized Jenkins for CI/CD.	
Software Tester Co-op Nokia Canada	Aug 2020 — Aug 2021 <i>Ottawa, ON</i>
- Enhanced JavaScript/XML code for better code coverage and robustness. - Produced design documents using Confluence Pages for system architecture and data flows.	

PROJECTS

pipi-shrimp-agent	github.com/mammut001/pipi-shrimp-agent
A high-performance desktop AI agent built with Tauri v2.	
Meet See U	chatpart-website.vercel.app
A full-stack AI social application featuring virtual companions with persistent vector memory and custom voice cloning.	
Focus Mint	apps.apple.com/us/app/focus-mint-focus-timer-study/id6759029810
A premium utility app for iOS, watchOS, and macOS designed to enhance productivity through Pomodoro-style timers and session tracking.	
System Optimization - Smart Path	system-optimizaion.vercel.app
A sophisticated algorithm visualization platform supporting graph theory (Dijkstra, A*, BFS/DFS), optimization (Simplex, Branch & Bound), and machine learning (Perceptrons).	
RentWise	github.com/random-char1198/Coin-Web-Crawler
Implemented Google SignUp and SignIn for user authentication, securely storing user info in Firebase.	

RESEARCH

Learning to See in the Dark 2018 Enhancement with DDBM	2024 — Present
Research on enhancing low-light image processing using DDBM (Denoising Diffusion Probabilistic Models) based on the foundational 'Learning to See in the Dark' paper.	

SKILLS

- Languages:** TypeScript, JavaScript, Python, Java, C++, Swift, Rust, HTML, CSS, Bash
- Frameworks & Data:** NextJS, TailwindCSS, Flask, Tauri v2, SQLite, MongoDB, JQuery
- Testing & Delivery:** PyTest, GTest, SeleniumGrid, Jenkins, Docker, Vagrant
- Tools & Platforms:** GitHub, Jira, Confluence, Linux, AWS, OpenStack, IntelliJ, PostMan, Windows10, Visual Studio, Typst, Mermaid, Object-Oriented Programming